FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – II
PAPER –1
TEST DATE: 04-01-2022

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-B & Section-C**.

Section – A ***(01 – 04, 20 – 23, 39 – 42)*****:** This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A ***(05 –10, 24 – 29, 43 – 48)*****:** This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B ***(11 – 13, 30 – 32, 49 – 51)*****:** This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.**

Section – C ***(14 – 19, 33 – 38, 52 – 57)*****:** This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

Physics

PART – I

Section – A (Maximum Marks: 12)

This section contains ***FOUR (04)*** *questions. Each question has* ***FOUR*** *options.* ***ONLY ONE*** *of these four options is the correct answer.*

1. There is a very small hole in a totally enclosed heated furnace. Outside the furnace, the air temperature is 0°C and the air pressure is 100kPa. The air inside the furnace is kept at a constant temperature of 57°C by the controlled heating system, and after a sufficiently long time its pressure also becomes stationary. Calculate (to the nearest integer) the magnitude of this stationary pressure.

(A) 105kPa

(B) 120kPa

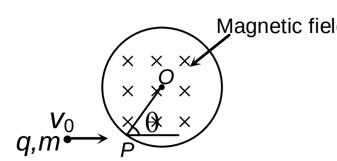
(C) 110kPa

(D) 124kPa

Ans. C

Sol. The number of molecules passing through a hole of cross-sectional area *A* during a time interval *t* is *A*v*Nt*, where v is the component, perpendicular to the wall, of the average molecular velocity, and *N* is the number density (the number of particles in unit volume). At equilibrium no of molecules going out should be equal to the number of molecules going in.
The square of the speed of the molecules (a measure of the internal energy of the gas) is proportional to the gas temperature *T*, and, from the ideal gas equation, the number density is proportional to the quotient of the pressure *p* and the tempera-ture.

2. A particle of charge *q* and mass m is projected with a velocity v_0 towards a circular region having uniform magnetic field *B* perpendicular and into the plane of paper from point *P* as shown in the figure. *R* is the radius and *O* is the centre of the circular region. If the line OP makes an angle θ with the direction of v_0 then the value of v_0 so that particle passes through *O* is



(A) $\frac{qBR}{m\sin\theta}$

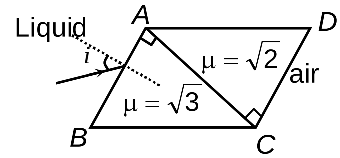
(B) $\frac{qBR}{2m\sin\theta}$

(C) $\frac{2qBR}{m\sin\theta}$

(D) $\frac{3qBR}{2m\sin\theta}$

Ans. B

3. A ray of light from a liquid $\left(\mu = \sqrt{3}\right)$ is incident on a system of two right angled prism of refractive indices $\sqrt{3}$ and $\sqrt{2}$ as shown. The ray suffers zero deviation when emerges into air from *CD*. The angle of incidence *i* is



(A) 45^0

(B) 35^0

(C) 20^0

(D) 10^0

Ans. A

4. In Young's experiment, the interference pattern is found to have an intensity ratio between the bright and dark fringes as 9. What is the ratio of intensities and amplitudes of the two interfering waves, respectively?

(A) $4, 2$

(B) $9, 3$

(C) $2.25, 1.5$

(D) None of these

Ans. A

Sol. In case of interference,

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$$

(a) for I to be maximum and minimum cos ϕ is 1 and – 1 respectively, i.e.,

$$I_{max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$$
$$= (\sqrt{I_1} + \sqrt{I_2})^2$$

and

$$I_{max} = I_1 + I_2 - 2\sqrt{I_1 I_2}$$
$$= \left(\sqrt{I_1} - \sqrt{I_2}\right)^2$$

According to given problem,

$$\frac{I_{max}}{I_{min}} = \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2} = \frac{9}{1},$$

i.e., $$\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_2} - \sqrt{I_1}} = \frac{3}{1}$$

By componendo and dividendo,

$$\frac{\sqrt{I_1}}{\sqrt{I_2}} = \frac{3+1}{3-1}$$

i.e., $$\frac{I_1}{I_2} = \frac{4}{1} = 4$$

(b) Now as for a wave I $\propto$ A^2,

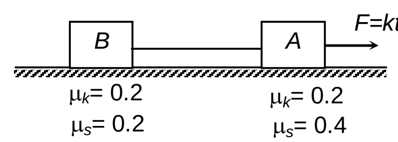
$$\frac{I_1}{I_2} = \left[\frac{A_1}{A_2}\right]^2, \left[\frac{A_1}{A_2}\right]^2 = 4$$

i.e., $$\frac{A_1}{A_2} = 2$$

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

5. Two blocks *A* and *B* each of mass $\frac{1}{2}$ kg is connected by a massless inextensible string and kept on horizontal surface. Coefficient of friction between block and surface is shown in figure. A force $F = kt$ (where *k* = 1 N/s and *t* is time in second) applied on *A*. Then (g = 10 m/s^2)



(A) work done by friction force on block *B* is zero in time interval *t* = 0 to *t* = 3s.

(B) work done by friction force on block *A* is zero in time interval *t* = 0 to *t* = 3s.

(C) work done by tension on *B* is also zero in time interval *t* = 0 to *t* = 3s.

(D) speed of blocks at *t* = 10s is 27.5 m/s.

Ans. ABC

6. A particle starts SHM at time $t = 0$. Its amplitude is *A* and angular frequency is ω. At time $t = 0$ its kinetic energy is $\frac{E}{4}$, where *E* is total energy. Assuming potential energy to be zero at mean position, the displacement-time equation of the particle can be written as

(A) $x = A\cos\left(\omega t + \frac{\pi}{6}\right)$

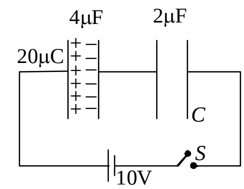
(B) $x = A\sin\left(\omega t + \frac{\pi}{3}\right)$

(C) $x = A\sin\left(\omega t - \frac{2\pi}{3}\right)$

(D) $x = A\cos\left(\omega t - \frac{\pi}{6}\right)$

Ans. ABCD

7. A $4\mu F$ capacitor is given $20\,\mu C$ charge and is connected with an uncharged capacitor of capacitance $2\,\mu F$ as shown in figure. When switch S is closed.



(A) charged flown through the battery is $\frac{40}{3}\mu C$

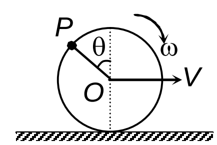
(B) charge flown through the battery is $\frac{20}{3}\mu C$

(C) work done by the battery is $\frac{200}{3}\mu J$

(D) work done by the battery is $\frac{100}{3}\mu J$

Ans. BC

8. A disc of radius R rolls on a horizontal surface with linear velocity V and angular velocity ω. There is a point P on circumference of disc at angle θ with upward vertical diameter measured anticlockwise (see figure), which has a vertical velocity. Here θ is equal to



(A) $\pi + \sin^{-1}\frac{V}{R\omega}$

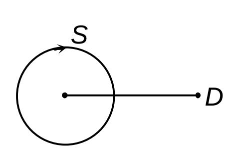
(B) $\frac{\pi}{2} - \sin^{-1}\frac{V}{R\omega}$

(C) $\pi - \cos^{-1}\frac{V}{R\omega}$

(D) $\pi + \cos^{-1}\frac{V}{R\omega}$

Ans. CD

9. A source of sound moves along a circle of radius 2 m with constant angular velocity 40 rad/s. Frequency of the source is 300 Hz. A detector is kept at some distance from the circle in the same plane of the circle (as shown in figure). Which of the following is not the possible value of frequency registered by the detector? (Speed of sound = 320 m/s)



(A) 250 Hz

(B) 360 Hz

(C) 410 Hz

(D) 220 Hz

Ans. CD

10. An electron in H-atom jumps from second excited state to first excited state and then from first excited to ground state. Let the ratio of wavelength, momentum and energy of photons emitted in these two cases be *a, b* and *c* respectively. Then

(A) $c = \frac{1}{a}$

(B) $a = \frac{9}{4}$

(C) $b = \frac{5}{27}$

(D) $c = \frac{5}{27}$

Ans. ACD

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

11. In a slow reaction, heat is being evolved at a rate about 10mW in a liquid. If the heat were being generated by the decay of ^{32}P, a radioactive isotope of phosphorus that has half-life of 14 days and emits only beta-particles with a mean energy of 700KeV. The number of ^{32}P atoms in the liquid is $\text{A}\times 77\times 10^{15}$ find a: [Take: $\ln 2$ = 0.7]

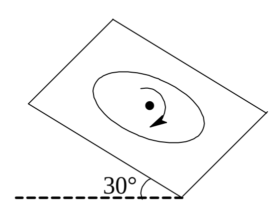
Ans. 2

Sol. $\text{P} = 700\times 10^{3}\times 1.6\times 10^{-19}\times \frac{\text{dn}}{\text{dt}} = 10\times 10^{-3}$

$$\frac{\text{dN}}{\text{dt}} = \frac{10^{-1}}{10^{-14}}\times \frac{1}{7\times 16} = \frac{10^{12}}{11.2} = \lambda\text{N}_0$$

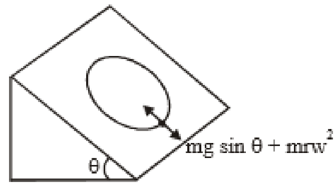
$$\lambda = \frac{\ln 2}{14\times 86400} \Rightarrow \text{N}_0 = \frac{14\times 86400\times 10^{12}}{11.2\ln 2} = 154\times 10^{15}$$

12. An old record player of 10 cm radius turns at 10 rad/s while mounted on a 30° incline as shown in the figure. A particle of mass m can be placed anywhere on the rotating record. If the least possible coefficient of friction μ that must exist for no slipping to occur is μ, find $2\sqrt{3}\mu$.



Ans. 6

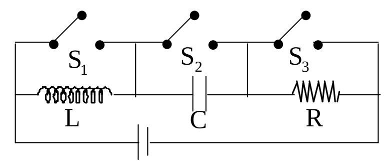
Sol. $\text{f} \geq \text{mg}\sin\theta + \text{mrw}^2$



$$\mu\, mg\cos\theta \geq mg\sin\theta + mrw^2$$

$$\text{or } \mu \geq \tan\theta + \frac{w^2 r}{g\cos\theta}$$

13. Consider the circuit shown in figure. With switch S_1 closed and the other two switches open, the circuit has a time constant 0.05 sec. With switch S_2 closed and the other two switches open, the circuit has a time constant 2 sec. With switch S_3 closed and the other two switches open, the circuit oscillates with a period T. Find T (in sec). (Take $\pi^2 = 10$)



Ans. 2

Sol. $\tau_1 = RC$

$$\tau_2 = \frac{L}{R}$$

$$\Rightarrow LC = t_1 t_2 = 0.1\text{sec}$$

$$\Rightarrow T = 2\pi\sqrt{LC} = 2\pi\sqrt{1/10} = 2\,\text{sec}$$

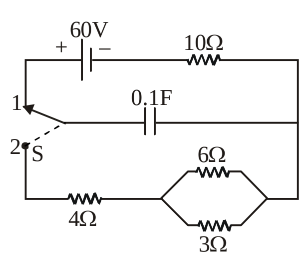
Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 14 and 15

Question Stem

A two way switch S is used in the circuit shown in fig. First, the capacitor is charged by putting the switch in position 1.



14. What is the heat generated across 4Ω resistor (in Joules)?

Ans. 120.00

15. What is the heat generated across 10Ω resistor (in Joules)?

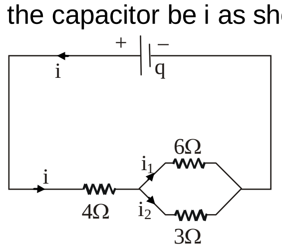
Ans. 0.00

Sol. Initially the switch was in position 1. Therefore, initially potential difference across capacitors was equal to emf of the battery i.e. 60 volt

$\therefore$ Initially energy stored in the capacitor was

$$U = \frac{1}{2}CV^2 = \frac{1}{2} \times 0.1 \times 60^2 \text{ J} = 180 \text{ J}$$

When switch is shifted to position 2, capacitor begins to discharge and energy stored in it is dissipated in the form of heat across resistances. Let at some instant discharging current through the capacitor be i as shown in fig.



According to Kirchoff's laws,

$i_1 + i_2 = i$ …(1)

$6i_1 - 3i_2 = 0$ or $i_2 = 2i_2$ …(2)

From above two equations,

$$i_1 = \frac{i}{3} \quad \text{and} \quad i_2 = \frac{2}{3}i$$

But thermal power generated in a resistance R is $P = i^2 R$ where is current flowing through it. Therefore, heat generated P_1, P_2 and P_3 across 4Ω, 6Ω and 3Ω resistances is in ratio

$4i^2 : 6i_1^2 : 3i_2^2$

or $P_1 : P_2 : P_3 = 4 : \frac{2}{3} : \frac{4}{3} = 6 : 1 : 2$

But total heat generated is

$P_1 + P_2 + P_3 = U$

$\therefore$ Heat generated across 4Ω is

$P_1 = 120$ J Ans.

Heat generated across 6Ω is

$P_2 = 20$ J Ans.

Heat generated across 3Ω is

$P_3 = 40$ J Ans.

Since, during discharging, no current flows through 10Ω, therefore heat generated across it is equal to zero.

Question Stem for Question Nos. 16 and 17

Question Stem

A choke coil is needed to operate an arc lamp at 160 V (rms) and 50 Hz. The arc lamp has an effective of 5 Ω when running at 10 A (rms). Compare the power losses in both cases.

16. Calculate the inductance of the choke coil (in H).

Ans. 0.05

17. If the same arc lamp is to be operated on 160V (dc), what additional resistance is required (in Ω)?

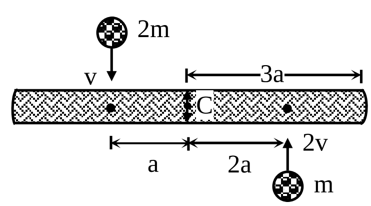
Ans. 11.00

Sol. Given that V_{rms} = 160 volt, f = 50 Hz.
R = Resistance of arc lamp = 5Ω.
i_{rms} = 10 amp = current through arc lamp.
If L be inductance of required choke coil then impedance of the circuit is
$Z = \sqrt{[R^2 + (2\pi fL)^2]} = \sqrt{[25 + (100\pi L)^2]}$
$\therefore$ current through arc lamp, $i_{rms} = \frac{V_{rms}}{Z}$
or $10 = \frac{160}{\sqrt{25+(100\pi L)^2}}$ or, $25 + (100\pi L)^2 = (16)^2 = 256$ or, $(100\pi L)^2$
$= 256 - 25 = 231$
$\therefore$ $L = \frac{\sqrt{231}}{100\pi}$ Henry = 0.05 H
When the same arc lamp is to be operated on 160 volt DC source, then
$\frac{160}{5+R}$ = i = 10. (where R is the additional resistance required to run the lamp)
Solving, we get : R = 11 Ω

Question Stem for Question Nos. 18 and 19

Question Stem

A uniform rod of mass 8m lies on a smooth horizontal table. Two point masses m and 2m moving in the same horizontal plane with speeds 2v and v (v = 10 m/s) respectively strike the bar as shown in Fig. and stick to the bar after collision. C denotes the center of mass of the rod and the length of the rod is 6a, where a = 1m.



18. What is the velocity of centre of mass just after collision (in m/s).

Ans. 0.00

19. What is the angular velocity of the system just after collision (in rad/s).

Ans. 2.00

Sol. Both the masses, 2m and m, after striking the bar give equal momentum each 2mv to the bar in opposite directions. Hence, after collision the bar has no translational motion, i.e. the linear velocity of the centre of mass of the bar is zero (v_c = 0)-

When both the masses stick to the bar, the whole system rotates about the centre of mass C. As there is no external torque acting on the system, the angular momentum is conserved.
Before collision (bar is stationary), there will be only the angular momentum of 2m and m about C. Hence the initial angular momentum (see Fig.) is
J_i = 2mva + m (2v) 2a = 6 m v a.
(in an anticlockwise direction)
After collision, the bar and both the masses (2m and m) rotate with angular velocity ω about the centre of mass C. The moment of inertia of the bar (mass 8m and length 6a) about C is
$\frac{M\ell^2}{12} = \frac{8m(6a)^2}{12} = 24m\,a^2$, and the moments of inertia of 2m and m about C are $2ma^2$ and $4ma^2$ respectively. Hence, after collision the angular momentum of the system is

$$J_f = (24ma^2 + 2ma^2 + 4ma^2)\,\omega = 30ma^2\omega$$

But $J_i = J_f$ (conservation of angular momentum)

$$\omega = \frac{6\,mav}{30\,ma^2} = \frac{v}{5a}$$

Chemistry

PART – II

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

20. The necessary condition for a compound to be optically active is:

(A) Presence of chiral carbon atom

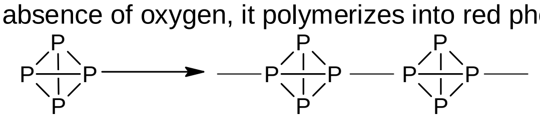
(B) Presence of atleast one asymmetric atom

(C) Non-superimposable mirror image.

(D) Absence of plane of symmetry.

Ans. C

21. White phosphorus is a tetra atomic solid $\left[P_4(s)\right]$ at room temperature and on strong heating in absence of oxygen, it polymerizes into red phosphorus as:



$\Delta H = -104$ KJ/mol of P_4

White(g) red(g)

The enthalpy of sublimation $\left[P_4(s) \longrightarrow P_4(g)\right]$ white is 59 KJ/mol and enthalpy of atomization is 316.25 KJ/mol of $P(g)$. The average $P-P$ bond enthalpy in P_4 molecule is:

(A) $102\,KJ$

(B) $201\,KJ$

(C) $104\,KJ$

(D) $120\,KJ$

Ans. B

22. Which of the following is weaker base?

(A) $N(SiH_3)_3$

(B) $N(CH_3)_3$

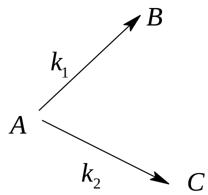
(C) $NH(CH_3)_3$

(D) All are equally basic

Ans. A

Sol. Due to back bonding (A) is a weaker base.

23. Consider the reaction



The rate constant for two parallel reactions were found to be $10^{-2}\ dm^3\ mol^{-1}\ s^{-1}$ and $4\times10^{-2} dm^3\ mol^{-1} s^{-1}$. If the corresponding energies of activation of the parallel reaction are 100 and 120 kJ/mol respectively, what is the net energy of activation (E_a) of A?

(A) $100 kJ / mol$

(B) $120 kJ / mol$

(C) $116 kJ / mol$

(D) $220 kJ / mol$

Ans. C

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

24. Which of the following have transition metal?

(A) Haemoglobin

(B) Vitamin B_{12}

(C) Cis – platin

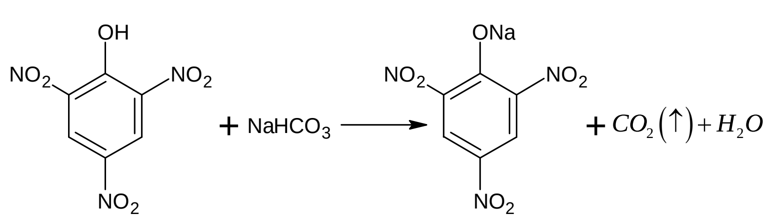
(D) Chlorophyll

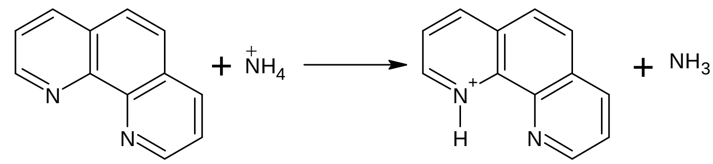
Ans. ABC

25. Which one of the following reactions is/are correct?

(A) $CH_3COOK + CF_3COOH \longrightarrow CH_3COOH + CF_3COOK$

(B)

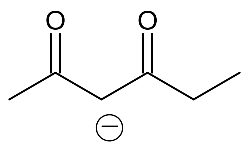


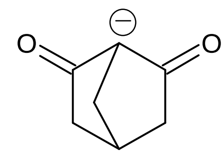
(C) 

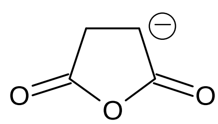
(D) $CH_3CH_2OH + NaHCO_3 \longrightarrow CH_3CH_3ONa + H_2O + CO_2\left(\uparrow\right)$

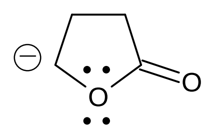
Ans. ABC

26. Which of the following carbanions are not resonance stabilized?

(A) 

(B) 

(C) 

(D) 

Ans. BD

27. Which of the following statements are correct?

(A) The highest probability of finding an electron in 1 s orbital is exactly at the middle between nucleus and circumference.

(B) In the Lyman series as the energy liberated during transition increases then distance between the special lines goes on decreasing.

(C) The highest probability of finding an electron in 1 s orbital is in the vicinity of the circumference.

(D) None of these

Ans. BC

28. Which of the following statements is / are correct?

(A) When lead nitrate is heated to at $400^{\circ}C$ only brown gas is released.

(B) Complete Hydrolysis of PCl_5 gives H_3PO_4 and HCl

(C) $N_2O_5(g) + 2NaOH_{(aq.)} \rightarrow 2NaNO_3 + H_2O$

(D) CCl_4 cannot be hydrolysed but hydrolysis of $SiCl_4$ conform $H_4SiO_4 + HCl$

Ans. BCD

29. Which of the following statements is (are) correct?

(A) Antimony on reaction with conc. HNO_3 gives antimonic acid.

(B) Manganese on reaction with cold and dilute HNO_3 gives NO_2 gas.

(C) HNO_2 disproportionate to give HNO_3 and NO

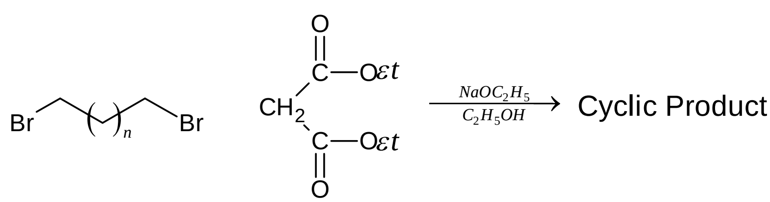
(D) HNO_3 on reaction with P_4O_{10} gives N_2O_5

Ans. ACD

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

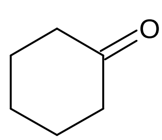
30.



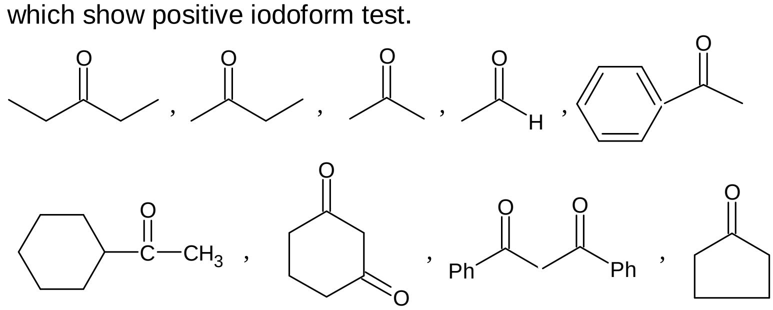
At what value of 'n' the formation of six membered ring take place.

Ans. 3

Sol. Product:

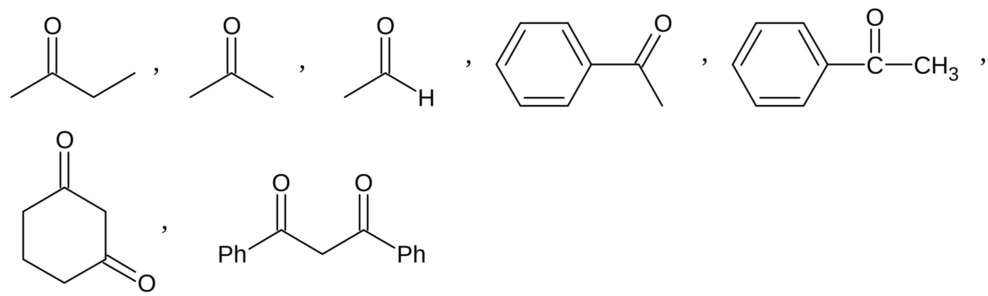


31. Examine the structural formulas of compounds given below and identify number of compounds which show positive iodoform test.



Ans. 6

Sol. These compounds will give iodoform test



32. For oxyacid $HClO_x$, if $x = y = z$ ($x, y \, and \, z$ are natural numbers), then calculate the value of $|x + y + z|$. Where $x =$ Number of 'O' atoms.

$\mathrm{y}=$ Total number of lone pairs at central atom

$\mathrm{z}=$ Total number of $\mathrm{pi}(\pi)$ electrons in the oxyacid

Ans. 6

Sol. The compound is $HClO_2$

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 33 and 34

Question Stem

The boiling point of water in a 0.1 molal silver nitrate solution (solution A) is x $°\mathrm{C}$. To this solution A, an equal volume of 0.1 molal aqueous barium chloride solution is added to make a new solution B. The difference in the boiling points of water in the two solutions A and B is $\mathrm{y} \times 10^{-2}\,°\mathrm{C}$.

(Assume: Densities of the solutions A and B are the same as that of water and the soluble salts dissociate completely.

Use: Molal elevation constant (Ebullioscopic Constant), $\mathrm{K_b}$ = 0.5 K kg $\mathrm{mol^{-1}}$; Boiling point of pure water as $100°C$.)

33. The value of x is___

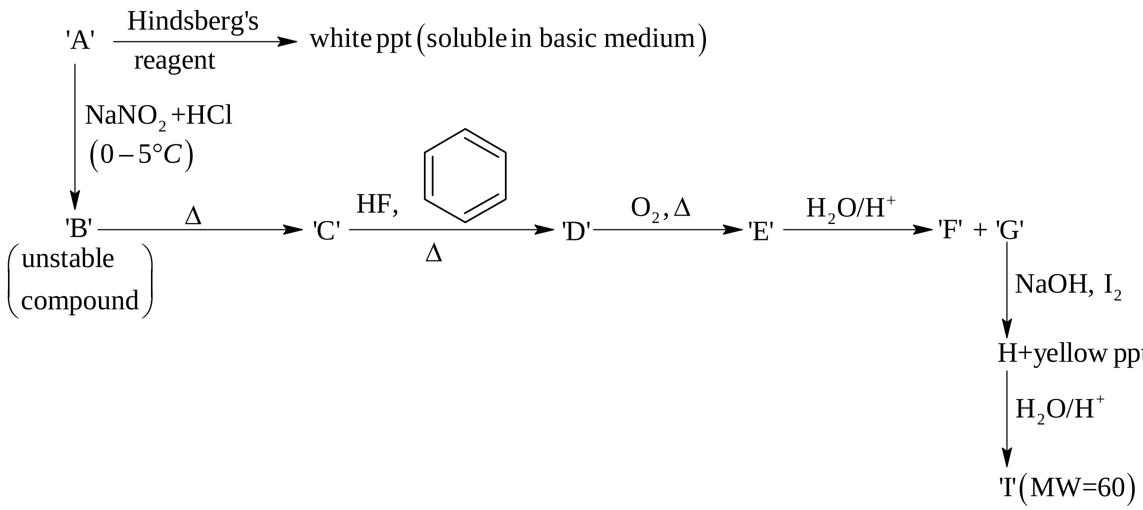
Ans. 100.10

34. The value of $|y|$ is____.

Ans. 2.50

Question Stem for Question Nos. 35 and 36

Question Stem



35. What is the degree of unsaturation in compound 'D'?

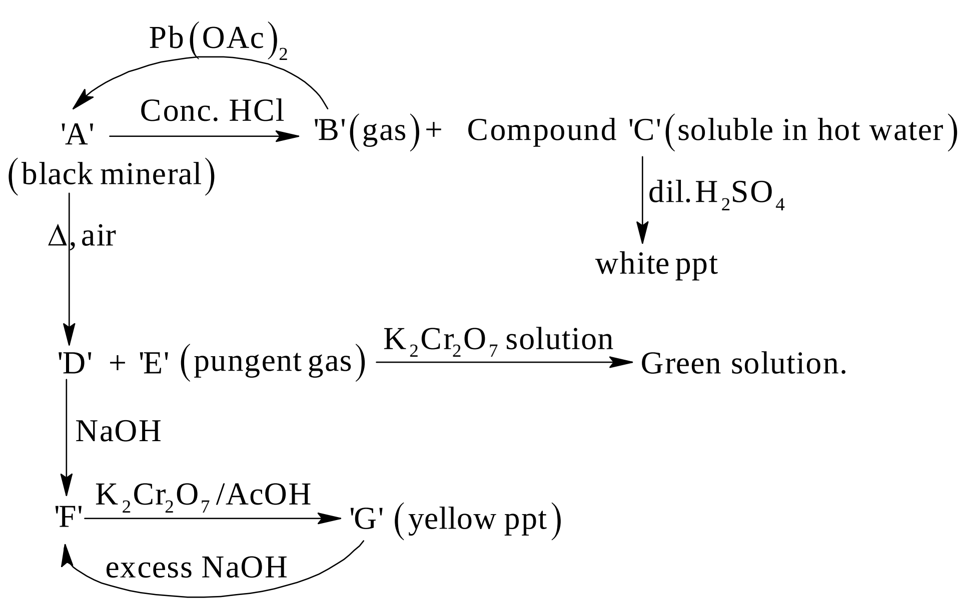
Ans. 4.00

36. What is the molecular weight of compound 'A'?

Ans. 59.00

Question Stem for Question Nos. 37 and 38

Question Stem



37. The number of $p\pi\text{-}d\pi$ bonds in the structure of gas $'E'$ is____.

Ans. 1.00

38. The sum of oxidation numbers of two metals present in yellow ppt $'G'$ is_____.

Ans. 8.00

Mathematics PART – III

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

39. If the normals at four points $(x_1, y_1),(x_2, y_2),(x_3, y_3)$ and (x_4, y_4) on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ are concurrent then $\sum \cos\alpha . \sum \sec\alpha$ is (where $\alpha, \beta,, \gamma, \delta$ are the eccentric angles of the points):

(A) 1

(B) 2

(C) 3

(D) 4

Ans. D

Sol. $(a^2-b^2)x^4-2ha^2(a^2-b^2)x^3-x^2+2a^4h(a^2-b^2)-a^6h^2=0 \Rightarrow \left(\sum x_1\right)\left(\sum \frac{1}{x_1}\right)=4$

40. Solution of differential equation $f'(x)f'''(x)=3\left(f''(x)\right)^2, \left(y=f(x)\right)$ is:

(A) $x=k_1y^2+k_2y+k_3$

(B) $y=k_1x^2+k_2+k_3$

(C) $x=\sin y$

(D) None of these

Ans. A

Sol. $\frac{3f''(x)}{f'(x)}=\frac{f'''(x)}{f''(x)}$

$\Rightarrow 3\ln f'(x)=\ln f''(x)+k$

$\Rightarrow \frac{d^2x}{dy^2}=c$

41. $f\left(x\right)=\begin{bmatrix} e^{-\sqrt{\left|\ln\{x\}\right|}}-\{x\}^{\sqrt{\frac{1}{\left|\ln\{x\}\right|}}} \\ 0 \qquad \text{otherwise} \end{bmatrix}$ Where ever it is defined is:

(A) not a periodic function

(B) even function

(C) range of $f\left(x\right)$ contains more than 1 element

(D) None of these

Ans. B

Sol. $e^{-\sqrt{-\ln t}}=t^{\frac{1}{\sqrt{-\ln t}}},t\in\left(0,1\right)$

42. Suppose f and g are functions having second derivatives f" and g" everywhere, if f(x) g(x) = 1 for all x and f ' (x) and g' are never zero, then $\frac{f''(x)}{f'(x)}-\frac{g''(x)}{g'(x)}$ equals

(A) $\frac{f'(x)}{f(x)}$

(B) $2\frac{f'(x)}{f(x)}$

(C) $-\frac{2g'(x)}{g(x)}$

(D) $-\frac{2f'(x)}{f(x)}$

Ans. B

Sol. $\frac{f''(x)}{f'\left(x\right)}-\frac{g''(x)}{g'\left(x\right)}=\frac{\frac{d}{dx}\left(f'(x)/g'(x)\right)}{\frac{f'\left(x\right).g'(x)}{\left(g'\left(x\right)\right)^2}}=\frac{g'}{f'}.\frac{d}{dx}\left(\frac{f'}{g'}\right)=\frac{2f'\left(x\right)}{f\left(x\right)}\left[\because f(x).g(x)=1\right].$

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

43. Let $f(x)=\begin{cases}\lim\limits_{n\to\infty}\left(\dfrac{px}{n}\sum\limits_{r=1}^{n}\dfrac{\left[r^2-e^{-x}+r-1\right]}{r(r+1)}\right)+\lambda, x>0\\ q, \qquad x=0\\ \lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\dfrac{\left\{r^2+r+e^x-1\right\}}{r(r+1)}, \quad x<0\end{cases}$

Is differentiable is R:

$([.]\text{is G.I.F.and}\{.\}\text{is F.P.of x})$ Then:

(A) $p=1$

(B) $q=1$

(C) $p+q+\lambda=3$

(D) if g is inverse of F then $g'(1/2)=2$

Ans. ABCD

Sol. $f(x)=\begin{cases}px+\lambda, x>0\\ q, x=0\\ e^x, x<0\end{cases}$

44. Consider $f(x)=4x^4-24x^3+31x^2+6x-8$ be a polynomial function and $\alpha,\beta,\gamma,\sigma$ are the roots. $(\alpha<\beta<\gamma<\sigma)$. So:

(A) $\sigma^p+\dfrac{1}{\sigma^\alpha}+\sigma^r+r^\sigma=36$

(B) $\sigma^p+\dfrac{1}{\sigma^\alpha}+\sigma^r+r^\sigma=18$

(C) $\displaystyle\int_{2\alpha}^{2\beta}\frac{x^{\sigma+1}-5x^{r+1}+2\beta|x|+1}{x^2+4\beta|x|+1}dx=2\ln 2$

(D) $\displaystyle\int_{2\alpha}^{2\beta}\frac{x^{\sigma+1}-5x^{r+1}+2\beta|x|+1}{x^2+4\beta|x|+1}dx=\ln 2$

Ans. AC

Sol. $f(x) = (2x-1)(2x+1)(x-2)(x-4)$

45. Each of 2010 boxes in a line contains one red marble and for $1 \le k \le 2010,$ the box is the k^{th} position also contain k white marbles. A child begins at the fist box and successively drawn a single marble at random from each box in order. He stops when be first draws a red marble. Let $p(n)$ be the probability that he stops after drawing exactly n marbles. The possible value(s) of n for which $p(n) < \frac{1}{2010}$ is:

(A) 44

(B) 45

(C) 46

(D) 47

Ans. BCD

Sol. $p(n) =$ Probability that child stops after drawing exactly n marbles

$$p(n) = \frac{1}{2} . \frac{2}{3} . \frac{3}{4} .... \left(\frac{n-1}{n}\right)\left(\frac{1}{n+1}\right) = \frac{1}{n(n+1)}$$

46. Consider the equation $z^2 - (3+i)z + (m+2i) = 0 \, (m \in R)$. If the equation has exactly one real and one-non-real complex root, then which of the following hold(s) good:

(A) Modulus of the non-real complex root is 2

(B) The value of m is 3

(C) Additive inverse of non-real root is $(-1-i)$

(D) Product of real root and imaginary part of non-real complex root is 2

Ans. CD

Sol. If α is real root then $\alpha = 2, m = \pm 2,$ non-real root = $1+i$

47. Consider a function $f(x) = \sin^{-1}\frac{2x}{1+x^2} + \cos^{-1}\frac{1-x^2}{1+x^2} + \tan^{-1}\frac{2x}{1-x^2} - a\tan^{-1}x \, (a \in R),$ the value of a if $f(x) = 0$ for all x:

(A) 6

(B) -6

(C) 2

(D) -2

Ans. ACD

Sol. If $0 < x < 1, a = 6;$ if $x > 1, a = 2,$ if $x < -1, a = -2$

48. $\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \frac{m^2 n}{\left(n.3^m + m.3^n\right)} = \frac{p}{q}$ (Where $p \,\&\, q$ are coprime) then:

(A) $p + q = 41$

(B) $p < q$

(C) p is perfect square

(D) q is perfect square

Ans. ABC

Sol. $2S = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \left(\frac{m.n}{m.3^n + n.3^n}\right)\left(\frac{n}{3^m} + \frac{m}{3^n}\right)$

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

49. $\lim_{n\to\infty}\left[\left((2009)^{2010}\right)^n + \left((2010)^{2009}\right)^n\right]^{1/n}$ is equal to $\left|a^b\right|$ where $a, b \in N$ then $a - b$ is....

Ans. 1

50. A and B are two square matrices such that $A^2 B = BA$ and if $(AB)^{10} = A^k B^{10}$ then the value of $k - 1020$ is....

Ans. 3

51. $AI_1I_2I_3$ is an excentral triangle of equilateral triangle ΔABC such that $I_1I_2 = 4$ unit, if ΔDEF is pedal triangle of ΔABC then $\sqrt{\frac{Ar(\Delta I_1I_2I_3)}{Ar(\Delta DEF)}} =$

Ans. 4

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 52 and 53

Question Stem

A plane P contains the lines $\mathrm{L}_1 : \frac{y}{b} + \frac{z}{c} = 1, x = 0$ and $\mathrm{L}_2 : \frac{x}{a} - \frac{z}{c} = 1, y = 0$

52. If the shortest distance between the lines L_1 and L_2 is $\frac{1}{4}$, then $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$ is______

Ans. 64.00

53. If B is the image of the point $A(1,0,0)$ in the plane $\mathrm{P}[\text{where a=b=c=1}]$ and $\mathrm{M}(1,2,3)$ is a point in the space then length BM is____

Ans. 2.24

Question Stem for Question Nos. 54 and 55

Question Stem

For the curves $\frac{x^2}{a^2} + \frac{y^2}{63} = 1$ and $y^2 = 4x$

54. The maximum integral value of a for which there is only one common normal to the two curves is_____.

Ans. 8.00

55. If the two curves have only two distinct common normal then the value of a is_____.

Ans. 9.00

Question Stem for Question Nos. 56 and 57

Question Stem

ABC is an isosceles triangle with AB = AC = 5 and BC = 6. Let P be a point inside the triangle ABC such that the distance from P to the base BC equals the geometric mean of the distances to the sides AB and AC.

56. The minimum distance of the point A from the locus of the point P is____.

Ans. 2.50

57. If R is a point of locus of point P which is nearest to the point A, then ratio of area of $\Delta\mathrm{ABC}$ to the area of $\Delta\mathrm{RBC}$ is_____.
[Answer should be rounded off to two decimal planes]

Ans. 2.67